Jiacheng Liang

814-470-0569 | ljcpro@outlook.com | LinkedIn | Google Scholar | jackpurcell.github.io

Research Interest

My research focuses on **LLM safety and trustworthy AI**, with an emphasis on identifying security vulnerabilities and developing robust defenses. Research areas include LLM post-training and safety alignment, LLM agent safety, LLM deployment security, jailbreak and backdoor defense, and RAG safety.

Education

Stony Brook University & Penn State University — Stony Brook, NY
Ph.D. Candidate in Computer Science, GPA: 4.0/4.0 — *Sep 2022 – Dec 2026 (Expected)*

University of Electronic Science and Technology of China — Chengdu, China
B.Eng. in Software Engineering – International Elite Class — *Sep 2018 – Jun 2022*

Experience

Amazon AGI Foundation – Nova Responsible AI Team — Boston, MA
Applied Scientist Intern — *May 2025 – Aug 2025*

- **Adaptive Red-Teaming for RLHF.** Identified systemic weaknesses in RLHF pipelines where the policy and reward model fail in tandem; proposed an adaptive red-teaming + end-to-end repair framework for more robust alignment. *(Pub. ARES, ACL '26)*

Publications

Advisor: Prof. Ting Wang (ALPS-Lab). Entries are first / co-first authored unless tagged **co-author**.

Attack & Red-Teaming

AutoRAN: Automated Hijacking of Safety Reasoning in Large Reasoning Models
ACL '26
Automatically hijacks internal safety reasoning in LRMs — turning chain-of-thought transparency into an attack surface.

ARES: Adaptive Red-Teaming and End-to-End Repair of Policy-Reward Systems
Amazon AGI · ACL '26
Exposes joint failures of policy & reward models in RLHF; proposes adaptive red-teaming + end-to-end repair.

GraphRAG under Fire
IEEE S&P '26 (Oakland)
First poisoning attack on GraphRAG — corrupts entity-relation graphs to hijack downstream knowledge retrieval.

Defense & Alignment

Data to Defense: The Role of Curation in Customizing LLMs Against Jailbreaking Attacks
EMNLP '25
End-to-end curation across LLM customization stages — achieves 100% safe-response rate against jailbreaks.

Robust Semantic Cache for LLM Serving
NeurIPS '26 sub.
Indexes cache on request + speculatively decoded tokens — provably resists cache poisoning while improving retrieval relevance.

RASA: Routing-Aware Safety Alignment for Mixture-of-Experts Models
COLM '26 sub.
Routing-aware bi-level alignment that robustifies both experts and routers in MoE LLMs against jailbreaks.

Dynamic Token Reweighting for Robust Vision-Language Models
CVPR '26 **co-author**
First KV-cache reweighting method for multimodal jailbreak defense — robustifies VLMs without retraining.

RobustKV: Defending Large Language Models against Jailbreak Attacks via KV Eviction
ICLR '25 co-author
Selectively evicts low-importance KV-cache tokens — neutralizes jailbreaks while preserving benign performance.

Your Agent Can Defend Itself Against Backdoor Attacks
Preprint co-author
Self-defense for LLM agents by enforcing consistency between planning, execution, and user instructions.

Benchmark & Evaluation

AgentLAB: Benchmarking LLM Agents Against Long-Horizon Attacks
ICML '26 co-author
First extensible benchmark for long-horizon attacks on LLM agents — uncovers vulnerabilities beyond single-turn defenses.

Model Extraction Attacks Revisited (MEBench)
AsiaCCS '24
Open-source platform assessing model-extraction vulnerabilities across real MLaaS APIs — unifies attacks, metrics, and victim models.

WaterPark: A Robustness Assessment of Language Model Watermarking
EMNLP '25
Systematization of LLM watermarking — standardized robustness eval, novel attack, and corresponding defenses.

Reasoning or Retrieval? A Study of Answer Attribution on Large Reasoning Models
ICLR '26 co-author
Diagnostic study attributing LRM answers between genuine reasoning and parametric retrieval.

Early Work

PASS: Patch Automatic Skip Scheme for Efficient On-Device Video Perception
IEEE TPAMI '24 co-author
Journal extension — adaptive patch-skip scheme for real-time on-device video perception.

PASS: Patch Automatic Skip Scheme for Efficient Real-Time Video Perception on Edge Devices
AAAI '23 co-author
Reduces redundant computation in video perception via learned patch-skip masks for edge devices.

Revisiting Frequency Analysis Against Encrypted Deduplication via Statistical Distribution
IEEE INFOCOM '22 co-author
Stronger frequency-analysis attack on encrypted deduplication storage.

Omnilytics: A Blockchain-based Secure Data Market for Decentralized ML
ICML FL Workshop '21
Blockchain-backed marketplace for secure decentralized ML data exchange.

Other Preprints

CyLens: Reinventing Cyber Threat Intelligence in the Paradigm of Agentic LLMs
Preprint
Agentic LLM framework for end-to-end cyber threat intelligence — ingestion, reasoning, and reporting.

Self-Improving Model Steering
Preprint co-author
Self-improvement loop for activation/representation steering of LLM behavior.

CopyrightMeter: Revisiting Copyright Protection in Text-to-Image Models
Preprint co-author
Unified evaluation of copyright protection mechanisms across text-to-image models.